

JESS Thermodynamic Database v8.9

Chemical species in reactions with Bismuth

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
[12]N4 [12]aneN4; 1,4,7,10-Tetraazacyclododecane; Cyclen; tcdd				
0	294-90-6	18	C(8)H(20)N(4)	172.274
[15]N4 [15]aneN4; 1,4,8,12-tetraazacyclopentadecane				
0	15439-16-4	21	C(11)H(26)N(4)	214.354
[9]N3:2OHEt*3 N,N',N''-Tris(2-hydroxyethyl)-1,4,7-triazacyclononane				
0	---	7	C(12)H(27)N(3)O(3)	261.365
222Dpa 2,2'-Ethylenebis(iminomethylene)dipyridine; Ethylenebis(iminomethylene-2-pyridine); N,N'-Di-2-picolylethylenediamine; EDDPY; 2,2,2-dpa; DPEN (sic); N,N'-Bis(2-pyridylmethyl)-1,2-diaminoethane; Ethylenebis(aminomethylene-2-pyridine)				
0	---	26	C(14)H(18)N(4)	242.324
2AmMePyridine 2-(Aminomethyl)pyridine; 2-Picolylamine; Aminomethylpyridine (2-); AMPY; 2-Aminomethylpyridine				
0	3731-51-9	57	C(6)H(8)N(2)	108.143
2SHEtol-1 2-Mercaptoethanol ion; Mercaptoethanol ion				
-1	---	66	C(2)H(5)O(1)S(1)	77.1211
2ThenTriFAcone-1				
-1	---	101	C(8)H(4)F(3)O(2)S(1)	221.174
8SHQuin-1				
-1	---	14	C(9)H(6)N(1)S(1)	160.213

As (s) Arsenic; Arsenic, rhombohedral; Arsenic, grey/gray; Arsenic, alpha				
0	7440-38-2	179	As (1)	74.9220
Ascorbic-2 L-Ascorbate ion; Ascorbate ion				
-2	63983-50-6	133	C (6) H (6) O (6)	174.110
Asn-1 Asparaginate ion; L-2-Aminobutanedioate 4-amide ion; L-beta-Asparaginate ion; Aspartate beta-amide ion; Altheinate ion; Asparamide ion; Agedoite ion				
-1	---	581	C (4) H (7) N (2) O (3)	131.111
AsO4-3 Arsenate ion				
-3	15584-04-0	238	As (1) O (4)	138.920
Asp-2 Aspartate ion; Aminosuccinate ion; Asparaginate ion; 2-Aminobutanedioate ion; L-2-Aminobutanedioate ion				
-2	63-71-8	776	C (4) H (5) N (1) O (4)	131.088
Bi+1				
1	---	2	Bi (1)	208.980
Bi+1_Cl-1_(s)				
0	---	1	Bi (1) Cl (1)	244.433
Bi+2 Bismuth(II) ion				
2	22341-84-0	7	Bi (1)	208.980
Bi+2_H+1_PAR-2				
1	---	1	C (11) H (8) Bi (1) N (3) O (2)	423.183
Bi+2_Thiourea				
2	---	1	C (1) H (4) Bi (1) N (2) S (1)	285.096
Bi+2_Thiourea (2)				

2	---	1	C(2)H(8)Bi(1)N(4)S(2)	361.212
Bi+2_Thiourea(3)				
2	---	1	C(3)H(12)Bi(1)N(6)S(3)	437.329
Bi+2_Thiourea(4)				
2	---	1	C(4)H(16)Bi(1)N(8)S(4)	513.445
Bi+2_Thiourea(5)				
2	---	1	C(5)H(20)Bi(1)N(10)S(5)	589.561
Bi+2_Thiourea(6)				
2	---	1	C(6)H(24)Bi(1)N(12)S(6)	665.677
Bi+3 Bismuth(III) ion				
3	23713-46-4	335	Bi(1)	208.980
Bi+3_[12]N4				
3	---	1	C(8)H(20)Bi(1)N(4)	381.254
Bi+3_[15]N4				
3	---	3	C(11)H(26)Bi(1)N(4)	423.334
Bi+3_[15]N4_OH-1				
2	---	1	C(11)H(27)Bi(1)N(4)O(1)	440.342
Bi+3_[9]N3:2OHEt*3				
3	---	1	C(12)H(27)Bi(1)N(3)O(3)	470.345
Bi+3_222Dpa				
3	---	1	C(14)H(18)Bi(1)N(4)	451.304
Bi+3_2AmMePyridine				
3	---	1	C(6)H(8)Bi(1)N(2)	317.123
Bi+3_2SHEtol-1				

2	---	1	C(2)H(5)Bi(1)O(1)S(1)	286.101
Bi+3_2SHEtol-1(2)				
1	---	1	C(4)H(10)Bi(1)O(2)S(2)	363.222
Bi+3_2SHEtol-1(3)				
0	---	1	C(6)H(15)Bi(1)O(3)S(3)	440.343
Bi+3_2ThenTriFAcane-1				
2	---	1	C(8)H(4)Bi(1)F(3)O(2)S(1)	430.154
Bi+3_2ThenTriFAcane-1(2)				
1	---	1	C(16)H(8)Bi(1)F(6)O(4)S(2)	651.328
Bi+3_2ThenTriFAcane-1(3)				
0	---	1	C(24)H(12)Bi(1)F(9)O(6)S(3)	872.501
Bi+3_8SHQuin-1				
2	---	2	C(9)H(6)Bi(1)N(1)S(1)	369.193
Bi+3_8SHQuin-1(2)				
1	---	1	C(18)H(12)Bi(1)N(2)S(2)	529.407
Bi+3_Ascorbic-2				
1	---	1	C(6)H(6)Bi(1)O(6)	383.090
Bi+3_Asn-1				
2	---	1	C(4)H(7)Bi(1)N(2)O(3)	340.091
Bi+3_Asn-1(2)				
1	---	1	C(8)H(14)Bi(1)N(4)O(6)	471.202
Bi+3_AsO4-3_(s) Bismuth arsenate				
0	---	2	As(1)Bi(1)O(4)	347.900
Bi+3_Asp-2				

1	---	2	C(4)H(5)Bi(1)N(1)O(4)	340.068
Bi+3_Asp-2(2)				
-1	---	3	C(8)H(10)Bi(1)N(2)O(8)	471.156
Bi+3_Asp-2(3)				
-3	---	1	C(12)H(15)Bi(1)N(3)O(12)	602.244
Bi+3_Bipy				
3	---	1	C(10)H(8)Bi(1)N(2)	365.167
Bi+3_Br-1				
2	---	3	Bi(1)Br(1)	288.884
Bi+3_Br-1_EDTA-4				
-2	---	2	C(10)H(12)Bi(1)Br(1)N(2)O(8)	577.098
Bi+3_Br-1(2)				
1	---	3	Bi(1)Br(2)	368.788
Bi+3_Br-1(3)				
0	---	2	Bi(1)Br(3)	448.692
Bi+3_Br-1(3)_(s) Bismuth tribromide; Bismuth bromide				
0	7787-58-8	2	Bi(1)Br(3)	448.692
Bi+3_Br-1(4)				
-1	---	2	Bi(1)Br(4)	528.596
Bi+3_Br-1(5)				
-2	---	2	Bi(1)Br(5)	608.500
Bi+3_Br-1(6)				
-3	---	2	Bi(1)Br(6)	688.404
Bi+3_BuXanthate-1(3)				

0	---	1	C(15)H(27)Bi(1)O(3)S(6)	656.718
Bi+3_CDTA-4				
-1	---	5	C(14)H(18)Bi(1)N(2)O(8)	551.286
Bi+3_Citric-3				
0	---	2	C(6)H(5)Bi(1)O(7)	398.082
Bi+3_Citric-3_(s)				
0	---	1	C(6)H(5)Bi(1)O(7)	398.082
Bi+3_Citric-3(2)				
-3	---	2	C(12)H(10)Bi(1)O(14)	587.183
Bi+3_Cl-1				
2	---	4	Bi(1)Cl(1)	244.433
Bi+3_Cl-1_Cys-2				
0	---	1	C(3)H(5)Bi(1)Cl(1)N(1)O(2)S(1)	363.571
Bi+3_Cl-1_I-1(2)				
0	---	2	Bi(1)Cl(1)I(2)	498.233
Bi+3_Cl-1_NO3-1				
1	---	1	Bi(1)Cl(1)N(1)O(3)	306.438
Bi+3_Cl-1_NO3-1(2)				
0	---	1	Bi(1)Cl(1)N(2)O(6)	368.443
Bi+3_Cl-1_NO3-1(3)				
-1	---	1	Bi(1)Cl(1)N(3)O(9)	430.448
Bi+3_Cl-1_OH-1(2)_(s)				
0	---	3	Bi(1)Cl(1)H(2)O(2)	278.448
Bi+3_Cl-1(2)				
1	---	5	Bi(1)Cl(2)	279.886

Bi+3_Cl-1(2)_I-1				
0	---	2	Bi(1)Cl(2)I(1)	406.786
Bi+3_Cl-1(2)_NO3-1				
0	---	1	Bi(1)Cl(2)N(1)O(3)	341.891
Bi+3_Cl-1(2)_NO3-1(2)				
-1	---	1	Bi(1)Cl(2)N(2)O(6)	403.896
Bi+3_Cl-1(2)_NO3-1(3)				
-2	---	1	Bi(1)Cl(2)N(3)O(9)	465.901
Bi+3_Cl-1(3)				
0	---	5	Bi(1)Cl(3)	315.339
Bi+3_Cl-1(3)_(g) Bismuth trichloride gas				
0	---	2	Bi(1)Cl(3)	315.339
Bi+3_Cl-1(3)_(s) Bismuth trichloride				
0	7787-60-2	2	Bi(1)Cl(3)	315.339
Bi+3_Cl-1(3)_NO3-1				
-1	---	1	Bi(1)Cl(3)N(1)O(3)	377.344
Bi+3_Cl-1(3)_NO3-1(2)				
-2	---	1	Bi(1)Cl(3)N(2)O(6)	439.349
Bi+3_Cl-1(4)				
-1	---	5	Bi(1)Cl(4)	350.792
Bi+3_Cl-1(5)				
-2	---	4	Bi(1)Cl(5)	386.245
Bi+3_Cl-1(6)				
-3	---	14	Bi(1)Cl(6)	421.698

Bi+3_Cs+1_Cl-1(6)				
-2	---	2	Bi(1)Cl(6)Cs(1)	554.608
Bi+3_Cs+1(2)_Cl-1(6)				
-1	---	2	Bi(1)Cl(6)Cs(2)	687.518
Bi+3_Cs+1(3)_Cl-1(6)				
0	---	2	Bi(1)Cl(6)Cs(3)	820.428
Bi+3_Cys-2				
1	---	1	C(3)H(5)Bi(1)N(1)O(2)S(1)	328.118
Bi+3_DDC-1(3)				
0	---	2	C(15)H(30)Bi(1)N(3)S(6)	653.763
Bi+3_Di2AmPyrid				
3	---	2	C(10)H(9)Bi(1)N(3)	380.182
Bi+3_Di2AmPyrid(2)				
3	---	1	C(20)H(18)Bi(1)N(6)	551.383
Bi+3_Dien				
3	---	3	C(4)H(13)Bi(1)N(3)	312.147
Bi+3_Dien_OH-1				
2	---	1	C(4)H(14)Bi(1)N(3)O(1)	329.155
Bi+3_Diglycol-2				
1	---	1	C(4)H(4)Bi(1)O(5)	341.053
Bi+3_Diglycol-2(2)				
-1	---	1	C(8)H(8)Bi(1)O(10)	473.126
Bi+3_Diglycol-2(3)				
-3	---	1	C(12)H(12)Bi(1)O(15)	605.198

Bi+3_DiMeDiSHCarbamic-1(3)				
0	---	1	C(9)H(18)Bi(1)N(3)S(6)	569.602
Bi+3_DiMePhenThiourea				
3	---	1	C(9)H(12)Bi(1)N(2)S(1)	389.248
Bi+3_Dithizone-1(3)				
0	---	2	C(39)H(33)Bi(1)N(12)S(3)	974.931
Bi+3_DTPA-5				
-2	---	4	C(14)H(18)Bi(1)N(3)O(10)	597.291
Bi+3_EDTA-4				
-1	---	11	C(10)H(12)Bi(1)N(2)O(8)	497.194
Bi+3_EGTA-4				
-1	---	1	C(14)H(20)Bi(1)N(2)O(10)	585.300
Bi+3_EtDiAm:2OHPr*4				
3	---	2	C(14)H(32)Bi(1)N(2)O(4)	501.399
Bi+3_EtDiAm:2OHPr*4_OH-1				
2	---	1	C(14)H(33)Bi(1)N(2)O(5)	518.406
Bi+3_EtThioUrea				
3	---	1	C(3)H(8)Bi(1)N(2)S(1)	313.150
Bi+3_F-1				
2	---	3	Bi(1)F(1)	227.978
Bi+3_F-1(2)				
1	---	4	Bi(1)F(2)	246.977
Bi+3_F-1(3)				
0	---	3	Bi(1)F(3)	265.975

Bi+3_F-1(3)_ (s) Bismuth fluoride; Bismuth trifluoride				
0	7787-61-3	2	Bi(1)F(3)	265.975
Bi+3_Fumaric-2				
1	---	1	C(4)H(2)Bi(1)O(4)	323.038
Bi+3_Gln-1				
2	---	1	C(5)H(9)Bi(1)N(2)O(3)	354.118
Bi+3_Gln-1(2)				
1	---	1	C(10)H(18)Bi(1)N(4)O(6)	499.256
Bi+3_Glu-2				
1	---	2	C(5)H(7)Bi(1)N(1)O(4)	354.095
Bi+3_Glu-2(2)				
-1	---	3	C(10)H(14)Bi(1)N(2)O(8)	499.210
Bi+3_Glu-2(3)				
-3	---	1	C(15)H(21)Bi(1)N(3)O(12)	644.325
Bi+3_Gly-1				
2	---	2	C(2)H(4)Bi(1)N(1)O(2)	283.039
Bi+3_Gly-1_OH-1				
1	---	2	C(2)H(5)Bi(1)N(1)O(3)	300.047
Bi+3_H+1_[15]N4				
4	---	1	C(11)H(27)Bi(1)N(4)	424.342
Bi+3_H+1_222Dpa				
4	---	1	C(14)H(19)Bi(1)N(4)	452.312
Bi+3_H+1_CDTA-4				
0	---	2	C(14)H(19)Bi(1)N(2)O(8)	552.294

Bi+3_H+1_Cl-1_Cys-2				
1	---	1	C(3)H(6)Bi(1)Cl(1)N(1)O(2)S(1)	364.579
Bi+3_H+1_Cys-2				
2	---	4	C(3)H(6)Bi(1)N(1)O(2)S(1)	329.126
Bi+3_H+1_Dien				
4	---	1	C(4)H(14)Bi(1)N(3)	313.155
Bi+3_H+1_DTPA-5				
-1	---	3	C(14)H(19)Bi(1)N(3)O(10)	598.299
Bi+3_H+1_EDTA-4				
0	---	3	C(10)H(13)Bi(1)N(2)O(8)	498.202
Bi+3_H+1_EGTA-4				
0	---	1	C(14)H(21)Bi(1)N(2)O(10)	586.308
Bi+3_H+1_Lys-1				
3	---	2	C(6)H(14)Bi(1)N(2)O(2)	355.169
Bi+3_H+1_Malic-2				
2	---	1	C(4)H(5)Bi(1)O(5)	342.061
Bi+3_H+1_NTA-3_Tiron-4				
-3	---	2	C(12)H(9)Bi(1)N(1)O(14)S(2)	664.302
Bi+3_H+1_Tartaric-2				
2	---	1	C(4)H(5)Bi(1)O(6)	358.060
Bi+3_H+1_Trien				
4	---	1	C(6)H(19)Bi(1)N(4)	356.224
Bi+3_H+1_TTHA-6				
-2	---	3	C(18)H(25)Bi(1)N(4)O(12)	698.396

Bi+3_H+1(2)_Ascorbic-2				
3	---	1	C(6)H(8)Bi(1)O(6)	385.106
Bi+3_H+1(2)_Cys-2(2)				
1	---	3	C(6)H(12)Bi(1)N(2)O(4)S(2)	449.272
Bi+3_H+1(2)_DTPA-5				
0	---	2	C(14)H(20)Bi(1)N(3)O(10)	599.307
Bi+3_H+1(2)_Lys-1(2)				
3	---	1	C(12)H(28)Bi(1)N(4)O(4)	501.359
Bi+3_H+1(2)_TTHA-6				
-1	---	2	C(18)H(26)Bi(1)N(4)O(12)	699.404
Bi+3_H+1(2)_XylenolOrange-6				
-1	---	1	C(31)H(28)Bi(1)N(2)O(13)S(1)	877.609
Bi+3_H+1(3)_MeThymolBlue-6				
0	---	3	C(37)H(41)Bi(1)N(2)O(13)S(1)	962.778
Bi+3_H+1(3)_TTHA-6				
0	---	3	C(18)H(27)Bi(1)N(4)O(12)	700.412
Bi+3_H+1(3)_XylenolOrange-6				
0	---	1	C(31)H(29)Bi(1)N(2)O(13)S(1)	878.617
Bi+3_H+1(4)_TTHA-6				
1	---	2	C(18)H(28)Bi(1)N(4)O(12)	701.420
Bi+3_H+1(5)_Malic-2(5)				
-2	---	1	C(20)H(25)Bi(1)O(25)	874.384
Bi+3_H+1(6)_XylenolOrange-6(2)				
-3	---	1	C(62)H(58)Bi(1)N(4)O(26)S(2)	1548.25

Bi+3_H+1(8)_MeThymolBlue-6(2)				
-1	---	1	C(74)H(84)Bi(1)N(4)O(26)S(2)	1718.59
Bi+3_HIMDA-2				
1	---	1	C(6)H(9)Bi(1)N(1)O(5)	384.121
Bi+3_HOEDTA-3				
0	---	4	C(10)H(15)Bi(1)N(2)O(7)	484.218
Bi+3_HydrazineNNDiAcet-2				
1	---	1	C(4)H(6)Bi(1)N(2)O(4)	355.083
Bi+3_I-1				
2	---	2	Bi(1)I(1)	335.880
Bi+3_I-1_EDTA-4				
-2	---	2	C(10)H(12)Bi(1)I(1)N(2)O(8)	624.094
Bi+3_I-1(2)				
1	---	1	Bi(1)I(2)	462.780
Bi+3_I-1(3)				
0	---	3	Bi(1)I(3)	589.680
Bi+3_I-1(3)_(s) Bismuth triiodide; Bismuth iodide				
0	7787-64-6	2	Bi(1)I(3)	589.680
Bi+3_I-1(4)				
-1	---	3	Bi(1)I(4)	716.580
Bi+3_I-1(5)				
-2	---	2	Bi(1)I(5)	843.480
Bi+3_I-1(6)				
-3	---	3	Bi(1)I(6)	970.380

Bi+3_I-1(7)				
-4	---	2	Bi(1)I(7)	1097.28
Bi+3_IDA-2				
1	---	1	C(4)H(5)Bi(1)N(1)O(4)	340.068
Bi+3_K+1_Cl-1(6)				
-2	---	2	Bi(1)Cl(6)K(1)	460.796
Bi+3_K+1(2)_Cl-1(6)				
-1	---	2	Bi(1)Cl(6)K(2)	499.894
Bi+3_K+1(3)_Cl-1(6)				
0	---	2	Bi(1)Cl(6)K(3)	538.992
Bi+3_Malic-2(3)				
-3	---	1	C(12)H(12)Bi(1)O(15)	605.198
Bi+3_MeThiourea				
3	---	1	C(2)H(6)Bi(1)N(2)S(1)	299.123
Bi+3_Na+1_Cl-1(6)				
-2	---	2	Bi(1)Cl(6)Na(1)	444.688
Bi+3_Na+1(2)_Cl-1(6)				
-1	---	2	Bi(1)Cl(6)Na(2)	467.678
Bi+3_NBuThiourea				
3	---	1	C(5)H(12)Bi(1)N(2)S(1)	341.204
Bi+3_NH3				
3	---	1	Bi(1)H(3)N(1)	226.011
Bi+3_NO3-1				
2	---	2	Bi(1)N(1)O(3)	270.985

Bi+3_NO3-1(2)				
1	---	3	Bi(1)N(2)O(6)	332.990
Bi+3_NO3-1(3)				
0	---	3	Bi(1)N(3)O(9)	394.995
Bi+3_NO3-1(4)				
-1	---	3	Bi(1)N(4)O(12)	457.000
Bi+3_NO3-1(5)				
-2	---	3	Bi(1)N(5)O(15)	519.005
Bi+3_NO3-1(6)				
-3	---	2	Bi(1)N(6)O(18)	581.009
Bi+3_NTA-3				
0	---	3	C(6)H(6)Bi(1)N(1)O(6)	397.097
Bi+3_NTA-3_Tiron-4				
-4	---	1	C(12)H(8)Bi(1)N(1)O(14)S(2)	663.294
Bi+3_NTA-3(2)				
-3	---	2	C(12)H(12)Bi(1)N(2)O(12)	585.214
Bi+3_OH-1				
2	---	5	Bi(1)H(1)O(1)	225.987
Bi+3_OH-1_Ascorbic-2				
0	---	1	C(6)H(7)Bi(1)O(7)	400.097
Bi+3_OH-1_CDTA-4				
-2	---	3	C(14)H(19)Bi(1)N(2)O(9)	568.293
Bi+3_OH-1_DMPS-3				
-1	---	1	C(3)H(6)Bi(1)O(4)S(3)	411.238

Bi+3_OH-1_DTPA-5				
-3	---	2	C(14)H(19)Bi(1)N(3)O(11)	614.298
Bi+3_OH-1_EDTA-4				
-2	---	3	C(10)H(13)Bi(1)N(2)O(9)	514.201
Bi+3_OH-1_HOEDTA-3				
-1	---	2	C(10)H(16)Bi(1)N(2)O(8)	501.226
Bi+3_OH-1(2)				
1	---	3	Bi(1)H(2)O(2)	242.995
Bi+3_OH-1(3)				
0	---	4	Bi(1)H(3)O(3)	260.002
Bi+3_OH-1(3)_(am.,s) Bismuth hydroxide, amorphous				
0	---	1	Bi(1)H(3)O(3)	260.002
Bi+3_OH-1(3)_(s) Bismuth hydroxide (hydrate)				
0	---	4	Bi(1)H(3)O(3)	260.002
Bi+3_OH-1(4)				
-1	---	5	Bi(1)H(4)O(4)	277.009
Bi+3_Oxalic-2				
1	---	2	C(2)Bi(1)O(4)	297.000
Bi+3_Oxalic-2(2)				
-1	---	2	C(4)Bi(1)O(8)	385.019
Bi+3_PO4-3				
0	---	1	Bi(1)O(4)P(1)	303.952
Bi+3_PO4-3_(s) Bismuth(III) phosphate				

0	---	1	Bi (1) O (4) P (1)	303.952
Bi+3_Rb+1_Cl-1 (6)				
-2	---	2	Bi (1) Cl (6) Rb (1)	507.166
Bi+3_Rb+1 (2)_Cl-1 (6)				
-1	---	2	Bi (1) Cl (6) Rb (2)	592.634
Bi+3_S2O3-2_EDTA-4				
-3	---	2	C (10) H (12) Bi (1) N (2) O (11) S (2)	609.312
Bi+3_SCN-1 Bismuth thiocyanate				
2	---	2	C (1) Bi (1) N (1) S (1)	267.058
Bi+3_SCN-1_EDTA-4				
-2	---	2	C (11) H (12) Bi (1) N (3) O (8) S (1)	555.272
Bi+3_SCN-1 (2)				
1	---	3	C (2) Bi (1) N (2) S (2)	325.135
Bi+3_SCN-1 (3)				
0	---	3	C (3) Bi (1) N (3) S (3)	383.213
Bi+3_SCN-1 (4)				
-1	---	3	C (4) Bi (1) N (4) S (4)	441.291
Bi+3_SCN-1 (5)				
-2	---	3	C (5) Bi (1) N (5) S (5)	499.369
Bi+3_SCN-1 (6)				
-3	---	2	C (6) Bi (1) N (6) S (6)	557.446
Bi+3_SO3-2_EDTA-4				
-3	---	2	C (10) H (12) Bi (1) N (2) O (11) S (1)	577.252
Bi+3_SO4-2				

1	---	1	Bi(1)O(4)S(1)	305.038
Bi+3_SO4-2(2)				
-1	---	1	Bi(1)O(8)S(2)	401.095
Bi+3_SO4-2(3)				
-3	---	1	Bi(1)O(12)S(3)	497.153
Bi+3_SO4-2(4)				
-5	---	1	Bi(1)O(16)S(4)	593.210
Bi+3_SO4-2(5)				
-7	---	1	Bi(1)O(20)S(5)	689.268
Bi+3_Succinic-2				
1	---	1	C(4)H(4)Bi(1)O(4)	325.053
Bi+3_Tartaric-2(2)				
-1	---	1	C(8)H(8)Bi(1)O(12)	505.124
Bi+3_Tet2PicEtDiAm				
3	---	1	C(26)H(28)Bi(1)N(6)	633.529
Bi+3_Tetren				
3	---	2	C(8)H(23)Bi(1)N(5)	398.284
Bi+3_Tetren_OH-1				
2	---	1	C(8)H(24)Bi(1)N(5)O(1)	415.291
Bi+3_Thiourea				
3	---	1	C(1)H(4)Bi(1)N(2)S(1)	285.096
Bi+3_Thiourea_EDTA-4				
-1	---	1	C(11)H(16)Bi(1)N(4)O(8)S(1)	573.310
Bi+3_Thiourea(2)				
3	---	1	C(2)H(8)Bi(1)N(4)S(2)	361.212

Bi+3_Thiourea(3)				
3	---	1	C(3)H(12)Bi(1)N(6)S(3)	437.329
Bi+3_Thiourea(6)				
3	---	1	C(6)H(24)Bi(1)N(12)S(6)	665.677
Bi+3_Trien				
3	---	2	C(6)H(18)Bi(1)N(4)	355.216
Bi+3_Trien_OH-1				
2	---	1	C(6)H(19)Bi(1)N(4)O(1)	372.223
Bi+3_TriEtOlAm				
3	---	2	C(6)H(15)Bi(1)N(1)O(3)	358.170
Bi+3_TriEtOlAm_OH-1				
2	---	1	C(6)H(16)Bi(1)N(1)O(4)	375.177
Bi+3_TriMeThiourea				
3	---	1	C(4)H(10)Bi(1)N(2)S(1)	327.177
Bi+3_TTHA-6				
-3	---	2	C(18)H(24)Bi(1)N(4)O(12)	697.388
Bi+3_XylenolOrange-6				
-3	---	1	C(31)H(26)Bi(1)N(2)O(13)S(1)	875.593
Bi+3(2)_OH-1(2)_CDTA-4(2)				
-4	---	1	C(28)H(38)Bi(2)N(4)O(18)	1136.59
Bi+3(2)_Oxalic-2(3)_(s)				
0	---	1	C(6)Bi(2)O(12)	682.019
Bi+3(2)_S-2(3)_(s) Bismuth sulphide				
0	1345-07-9	3	Bi(2)S(3)	514.140

Bi+3 (2) _Se-2 (3) _ (s) Bismuth selenide				
0	---	1	Bi (2) Se (3)	654.849
Bi+3 (2) _SO4-2 (3) _ (s) Bismuth sulfate				
0	---	2	Bi (2) O (12) S (3)	706.133
Bi+3 (2) _Te-2 (3) _ (s) Bismuth telluride				
0	---	1	Bi (2) Te (3)	800.769
Bi+3 (2) _XylenolOrange-6 (2)				
-6	---	1	C (62) H (52) Bi (2) N (4) O (26) S (2)	1751.19
Bi+3 (6) _OH-1 (12)				
6	---	6	Bi (6) H (12) O (12)	1457.97
Bi+3 (9) _OH-1 (20)				
7	---	6	Bi (9) H (20) O (20)	2220.97
Bi+3 (9) _OH-1 (21)				
6	---	6	Bi (9) H (21) O (21)	2237.97
Bi+3 (9) _OH-1 (22)				
5	---	5	Bi (9) H (22) O (22)	2254.98
Bi+5				
5	---	1	Bi (1)	208.980
Bi (g)				
0	---	1	Bi (1)	208.980
Bi (s) Bismuth; Bismuth, rhombohedral				
0	7440-69-9	61	Bi (1)	208.980

Bi₂O₃(s) Bismuth trioxide; Bismuth oxide, alpha; Bismite; Bismuth oxide, yellow monoclinic				
0	1304-76-3	6	Bi(2)O(3)	465.958
Bi₂O₄(s) Bismuth tetroxide				
0	12048-50-9	1	Bi(2)O(4)	481.958
Bi₂O₅(s) Bismuth pentoxide				
0	---	1	Bi(2)O(5)	497.957
Bi₆O₆+6				
6	---	1	Bi(6)O(6)	1349.88
Bi₆O₆+6_OH-1(3)				
3	---	1	Bi(6)H(3)O(9)	1400.90
BiH₃(g) Bismuth hydride				
0	18288-22-7	1	Bi(1)H(3)	212.004
BiI(s) Bismuth monoiodide				
0	---	1	Bi(1)I(1)	335.880
BiO+1				
1	---	8	Bi(1)O(1)	224.979
BiO+1(2)_CO₃-2_(s) Bisumtite				
0	---	1	C(1)Bi(2)O(5)	509.968
BiOBr(s) Bromooxobismuthine				
0	7787-57-7	3	Bi(1)Br(1)O(1)	304.883
BiOCl(s) Bismoclite; Chlorooxobismuthine; Bismuth oxychloride; Bismuth subchloride				

0	7787-59-9	5	Bi(1)Cl(1)O(1)	260.432
BiONO3(s) Bismuth oxide nitrate				
0	10361-46-3	2	Bi(1)N(1)O(4)	286.984
BiOOH(s) Bismuth hydroxide				
0	---	2	Bi(1)H(1)O(2)	241.987
Bipy 2,2'-Bipyridyl; Bipyridine; 2,2'-Bipyridine; Dipyridyl; 2,2'-Dipyridyl; 2-(2-Pyridyl)pyridine; bipy; bpy; dip; dipy; alpha-alpha'-Bipyridyl				
0	366-18-7	823	C(10)H(8)N(2)	156.187
Br-1 Bromide ion				
-1	24959-67-9	1213	Br(1)	79.9040
Br2(l) Bromine, liquid				
0	---	285	Br(2)	159.808
BuXanthate-1				
-1	---	14	C(5)H(9)O(1)S(2)	149.246
C(s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C(1)	12.0110
CDTA-4 trans-1,2-cyclohexylenedinitrilotetraacetate ion; trans-1,2-diaminocyclohexanetetraacetate ion; CDTA ion				
-4	22005-54-5	301	C(14)H(18)N(2)O(8)	342.306
Citric-3 Citrate ion				
-3	126-44-3	1347	C(6)H(5)O(7)	189.102
Cl-1 Chloride ion				

-1	16887-00-6	2242	Cl (1)	35.4530
Cl2 (g) Chlorine gas				
0	7782-50-5	589	Cl (2)	70.9060
Cs+1 Caesium(I) ion; Cesium(I) ion				
1	18459-37-5	246	Cs (1)	132.910
Cys-2 Cysteinate ion; L-Cysteinate ion; beta-mercaptoalanate ion; 2-amino-3-mercaptopropanoate ion; 2-amino-3-mercaptopropionate ion; alpha-amino-beta-thiolpropionate ion				
-2	104170-20-9	673	C (3) H (5) N (1) O (2) S (1)	119.138
DDC-1				
-1	---	43	C (5) H (10) N (1) S (2)	148.261
Di2AmPyrid 2,2'-Dipyridylamine; Bis-(2-pyridyl)amine; Iminodi-2-pyridine; Di-2-pyridylamine				
0	1202-34-2	36	C (10) H (9) N (3)	171.202
Dien Diethylenetriamine; Iminobis(2-ethylamine); Dien; 1,4,7-Triazaheptane; 3-Azapentane-1,5-diamine; N1-(2-Aminoethyl)-1,2-ethanediamine; 3NH-pd; 2,2-tri; 2,2'-Diaminodiethylamine; 2,2'-Iminobis(ethylamine)				
0	111-40-0	254	C (4) H (13) N (3)	103.167
Diglycol-2 Diglycolate ion				
-2	---	130	C (4) H (4) O (5)	132.073
DiMeDiSHCarbamic-1				
-1	---	24	C (3) H (6) N (1) S (2)	120.207
DiMePhenThiourea Dimethylphenylthiourea; N,N'-Dimethyl-N-phenylthiourea				
0	---	1	C (9) H (12) N (2) S (1)	180.268
Dithizone-1				

-1	---	74	C(13)H(11)N(4)S(1)	255.317
DMPS-3 2,3-dimercaptopropane-1-sulphonate ion; unithiolate ion				
-3	---	48	C(3)H(5)O(3)S(3)	185.251
DTPA-5 Diethylenetrinitrilotetraacetate ion; 1,4,7-triazaheptane-1,1,7,7-pentaacetate ion; N,N-bis[2-[bis(carboxymethyl)amino]ethyl]glycinate ion; DTPA ion				
-5	---	343	C(14)H(18)N(3)O(10)	388.311
e-1 Electron				
-1	---	3399	E(1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C(10)H(12)N(2)O(8)	288.214
EGTA-4 Ethylenebis(oxyethylenenitrilo)tetraacetate ion; EGTA ion				
-4	370-65-0	132	C(14)H(20)N(2)O(10)	376.320
EtDiAm:2OHPr*4 N,N,N',N'-Tetrakis(2-hydroxypropyl)ethylenediamine; DL-Ethylenedinitrilotetra-1-(2-propanol); Ethylenedinitrilotetra-1-(2-propanol); Tetrakis(2-hydroxypropyl)ethylenediamine; THPEN				
0	102-60-3	19	C(14)H(32)N(2)O(4)	292.419
EtThioUrea Ethylthiourea; N-Ethylthiourea				
0	625-53-6	15	C(3)H(8)N(2)S(1)	104.170
F-1 Fluoride ion				
-1	16984-48-8	1090	F(1)	18.9984
F2(g) Fluorine gas				
0	7782-41-4	259	F(2)	37.9968

Fumaric-2 Fumarate ion; trans-Butenedioate ion				
-2	142-42-7	70	C(4)H(2)O(4)	114.058
Gln-1 Glutamate ion; L-Glutamate ion; 2-aminoglutaramate ion; glutamate 5-amide ion; L-2-aminopentanedioate 5-amide ion				
-1	---	536	C(5)H(9)N(2)O(3)	145.138
Glu-2 Glutamate ion; 2-aminopentanedioate ion; L-Glutamate ion; alpha-aminoglutarate ion; 1-aminopropane-1,3-dicarboxylate ion				
-2	138-18-1	650	C(5)H(7)N(1)O(4)	145.115
Gly-1 Glycinate ion; Aminoacetate ion				
-1	---	1356	C(2)H(4)N(1)O(2)	74.0593
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H(1)	1.00794
H+1_BiO3-1				
0	---	1	Bi(1)H(1)O(3)	257.986
H+1_CDTA-4				
-3	---	39	C(14)H(19)N(2)O(8)	343.313
H+1_Cys-2				
-1	---	39	C(3)H(6)N(1)O(2)S(1)	120.146
H+1_DDC-1 Diethyldithiocarbamic acid; Dithiocarb; Diethylcarbamodithioic acid; DTC (Dithiocarb); DDC; DEDC; DDTC; DeDTC				
0	---	15	C(5)H(11)N(1)S(2)	149.269
H+1_Dithizone-1 Dithizone; Diphenylthiocarbazone; 1,5-Diphenyl-1,2,4,5-tetraazapent-1-en-3-thione; 1-Phenyl-4-(phenylimino)-3-thio-semicarbazide; 1,5-Diphenyl-3-formazathiol; Dithizon (no 'e')				
0	60-10-6	30	C(13)H(12)N(4)S(1)	256.325

H+1_DTPA-5				
-4	---	44	C(14)H(19)N(3)O(10)	389.319
H+1_EDTA-4				
-3	---	70	C(10)H(13)N(2)O(8)	289.222
H+1_EGTA-4				
-3	---	22	C(14)H(21)N(2)O(10)	377.328
H+1_F-1				
0	---	142	F(1)H(1)	20.0063
H+1_Lys-1 Lysine; L-Lysine; 2,6-Diaminohexanoic acid; alpha,epsilon-Diaminocaproic acid				
0	---	59	C(6)H(14)N(2)O(2)	146.189
H+1_Malic-2				
-1	---	24	C(4)H(5)O(5)	133.081
H+1_PAR-2				
-1	---	46	C(11)H(8)N(3)O(2)	214.203
H+1_Tartaric-2				
-1	---	39	C(4)H(5)O(6)	149.080
H+1(2)_Cys-2 Cysteine; L-Cysteine; (R)-(+)-Cysteine; beta-Mercaptoalanine; 2-Amino-3-mercaptopropanoic acid				
0	52-90-4	13	C(3)H(7)N(1)O(2)S(1)	121.154
H+1(2)_DTPA-5				
-3	---	16	C(14)H(20)N(3)O(10)	390.327
H+1(2)_S-2 Hydrogen sulphide; Hydrogen sulfide				
0	---	58	H(2)S(1)	34.0759
H+1(2)_Tiron-4				

-2	---	27	C(6)H(4)O(8)S(2)	268.213
H+1(2)_XylenolOrange-6				
-4	---	23	C(31)H(28)N(2)O(13)S(1)	668.629
H+1(3)_Cys-2				
1	---	7	C(3)H(8)N(1)O(2)S(1)	122.162
H+1(3)_MeThymolBlue-6				
-3	---	28	C(37)H(41)N(2)O(13)S(1)	753.798
H+1(3)_XylenolOrange-6				
-3	---	11	C(31)H(29)N(2)O(13)S(1)	669.637
H+1(5)_MeThymolBlue-6				
-1	---	7	C(37)H(43)N(2)O(13)S(1)	755.814
H+1(9)_TTHA-6				
3	---	1	C(18)H(33)N(4)O(12)	497.480
H2(g) Hydrogen gas; Hydrogen				
0	1333-74-0	1564	H(2)	2.01588
H2O Water				
0	7732-18-5	5947	H(2)O(1)	18.0153
HIMDA-2				
-2	---	201	C(6)H(9)N(1)O(5)	175.141
HOEDTA-3 N-[2-[bis(carboxymethyl)amino]ethyl]-N-(2-hydroxyethyl)glycinate ion; N-(2-Hydroxyethyl)ethylenedinitrilotriacetate; HEDTA-3				
-3	14047-40-6	321	C(10)H(15)N(2)O(7)	275.238
HydrazineNNDiAcet-2 Hydrazine-N,N'-diacetate anion				
-2	---	80	C(4)H(6)N(2)O(4)	146.103

I-1 Iodide ion				
-1	20461-54-5	894	I (1)	126.900
I2 (s) Iodine; Iodine, orthorhombic				
0	7553-56-2	244	I (2)	253.800
IDA-2 Iminodiacetate ion				
-2	28528-43-0	428	C (4) H (5) N (1) O (4)	131.088
K+1 Potassium(I) ion				
1	24203-36-9	462	K (1)	39.0980
Malic-2 Malate ion				
-2	149-61-1	682	C (4) H (4) O (5)	132.073
MeThiourea Methylthiourea; N-Methylthiourea				
0	598-52-7	16	C (2) H (6) N (2) S (1)	90.1430
MeThymolBlue-6				
-6	---	114	C (37) H (38) N (2) O (13) S (1)	750.774
Na+1 Sodium(I) ion				
1	17341-25-2	617	Na (1)	22.9900
NBuThiourea N-Butylthiourea				
0	---	6	C (5) H (12) N (2) S (1)	132.224
NH3 Ammonia, aqueous				
0	---	1462	H (3) N (1)	17.0305

NO3-1 Nitrate ion				
-1	14797-55-8	573	N(1)O(3)	62.0049
NTA-3 Nitrilotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion; Triglycinate ion; NTA ion				
-3	28528-44-1	751	C(6)H(6)N(1)O(6)	188.117
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C(2)O(4)	88.0196
Pb (s) Lead; Lead, cubic				
0	7439-92-1	127	Pb(1)	207.200
PbBiO2Cl (s) Perite				
0	---	1	Bi(1)Cl(1)O(2)Pb(1)	483.632
PO4-3 Phosphate ion; Phosphate ion				
-3	14265-44-2	1594	O(4)P(1)	94.9716
Rb+1 Rubidium(I) ion				
1	22537-38-8	125	Rb(1)	85.4680
S-2 Sulfide ion; Sulphide ion				
-2	14127-58-3	466	S(1)	32.0600

S (s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S (1)	32.0600
S2O3-2 Thiosulphate ion; Thiosulfate ion				
-2	14383-50-7	276	O (3) S (2)	112.118
SCN-1 Thiocyanate ion; NCS-1 equivalent				
-1	302-04-5	785	C (1) N (1) S (1)	58.0777
Se (s) Selenium, trigonal; Selenium, grey; Selenium, black; Selenium, gamma; Selenium, hexagonal (sic)				
0	7782-49-2	140	Se (1)	78.9630
SO3-2 Sulphite ion; Sulfite ion				
-2	14265-45-3	146	O (3) S (1)	80.0582
SO4-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O (4) S (1)	96.0576
Succinic-2 Succinate ion				
-2	56-14-4	635	C (4) H (4) O (4)	116.073
Tartaric-2 Tartrate ion				
-2	---	393	C (4) H (4) O (6)	148.072
Te (s) Tellurium; Tellurium, hexagonal				
0	13494-80-9	71	Te (1)	127.603
Tet2PicEtDiAm 2,2',2'',2'''-(Ethylenedinitrilo)tetramethylenetetrapyridine; Ethylenedinitrilotetrakis(methylene-2-pyridine); N,N,N',N'-Tetra-2-				

picolythylenediamine; TPEN; EDTPY; N,N,N',N'-Tetrakis(2-pyridylmethyl)ethylenediamine				
0	---	25	C(26)H(28)N(6)	424.549
Tetren Tetraethylenepentamine; 1,4,7,10,13-Penta-azatridecane; TEPA; 2,2,2,2-pent; 3,6,9-Triazaundecane-1,11-diamine; Tetren; N1-(2-Aminoethyl)-N2-[2-[(2-aminoethyl)amino]ethyl]-1,2-ethanediamine				
0	112-57-2	93	C(8)H(23)N(5)	189.304
Thiourea Thiocarbamide; Thiourea; tu				
0	62-56-6	261	C(1)H(4)N(2)S(1)	76.1162
Trien Triethylenetetramine; 1,4,7,10-Tetraazadecane; N,N'-bis(2-aminoethyl)-1,2-ethanediamine; Trien; 2,2,2-tet; 3,6-Diazaoctane-1,8-diamine; Ethylenebis(2'-aminoethylamine); TETA (Triethylenetetramine); 3,6NH-od; 1,2-Diamino-N,N'-di(2-aminoethyl)ethane; N1,N2-Bis(2-aminoethyl)-1,2-ethanediamine				
0	112-24-3	155	C(6)H(18)N(4)	146.236
TriEtOlAm Triethanolamine; Tris(2-hydroxyethyl)amine; 2,2',2''-Nitrilotriethanol; Tea; H3tea				
0	102-71-6	84	C(6)H(15)N(1)O(3)	149.190
TriMeThiourea Trimethylthiourea; N,N,N'-Trimethylthiourea				
0	---	2	C(4)H(10)N(2)S(1)	118.197
TTHA-6 Triethylenetetranitrihexa-acetate				
-6	---	266	C(18)H(24)N(4)O(12)	488.408
XylenolOrange-6				
-6	---	60	C(31)H(26)N(2)O(13)S(1)	666.613